Design Methods in Practice

Amy J. Ko, *Design Methods*, 2nd edition. Prepared for Major Studio, MS Data Viz.
Case: DVC's first career panel. Spoken notes below target 4:45, with a 15-second buffer.

0:00-0:40 | Design is a way of reasoning [1-2]

Amy J. Ko's Design Methods presents design as a practice of generating, communicating, and evaluating ideas. It reaches beyond visual appearance. The book introduces methods for understanding a situation, exploring possibilities, and learning from what happens. I am applying those methods to DVC's first career panel. Our goal today is to practice design reasoning; we do not need to finish an event plan.

0:40-1:25 | Understand people before choosing a solution [3-4]

A career panel is already a proposed solution. First, we need to identify whose situation we want to improve. Someone comparing careers may need different support from someone seeking a first internship. Interviews and observation can help us understand those experiences. We then synthesize what we learn into goals, personas, scenarios, and a clear problem argument. These representations need grounding in research. Our classroom vote will select a hypothesis to investigate, rather than establish what all attendees need.

1:25-2:05 | Generate alternatives and make them visible [5]

Creativity involves practice, context, and a willingness to explore. We can ask people for ideas, borrow from other settings, question assumptions, and externalize possibilities. For DVC, we might compare shared questions across speakers, individual pathway stories, and anonymous Q&A.; Each format could support a different need and introduce a different tradeoff. Making alternatives visible gives us something specific to compare before committing to our first idea.

2:05-2:55 | Prototype to answer a question [6-7]

A prototype should help us learn something that affects a decision. We choose its detail around that uncertainty. Our question form explores whether someone can contribute without speaking to the room and understand what happened afterward. Typing and submitting provide input; draft and submitted are states; the confirmation provides feedback. This is relevant to interactive visualization: a filter or selection also changes state, and the display must help people interpret the result.

2:55-4:00 | Distinguish three ways of evaluating [8-10]

Critique examines reasoning through dialogue: someone explains a choice, and others question its goals, assumptions, and consequences. Empirical evaluation observes people using a design. Analytical evaluation inspects a design using principles or a structured walkthrough to predict difficulties. These methods offer different kinds of evidence. A participant completing our form can reveal something about that interaction, but it cannot establish whether the career panel provides useful advice. That broader claim needs investigation in a relevant setting with the intended attendees.

4:00-4:45 | Iteration and the studio connection [2]

The relationships between these methods matter more than treating chapters as a checklist. Feedback may change an interface, suggest another format, or make us redefine the problem. Participation, values, and power also shape whose ideas count. For our visualization practice, the takeaway is to connect audience, purpose, representation, interaction, and evidence. We will now make three decisions and keep track of the assumptions we still need to investigate.

Three questions for the room

Use these five-minute blocks after the summary. The existing app's tab timings and G guide describe its original compact 15-minute route; this outline supplies the assignment's 5 + 15-minute structure.

01 / 5:00-10:00 / CHAPTERS 2-4

Whose problem are we solving, and what evidence might change our mind?

Open: The case + The problem. Name a specific group of intended attendees. Choose a working need: comparing careers, finding a first step, asking candid questions, or another need. Discuss a concrete experience we would need to learn about from those people.

Facilitate: 0:30 name the group; 1:00 choose a hypothesis; 2:30 discuss evidence and missing perspectives; 1:00 record the reasoning.

Optional probes: Ask what finding would challenge the choice, who is absent from this classroom, and whether an interview, observation, or another source would help. **Record:** One audience, one need, and one piece of evidence to seek. Keep disagreement visible.

02 / 10:00-15:00 / CHAPTERS 5-8

Which panel format fits that need, and what tradeoff are we willing to accept?

Open: Alternatives + A prototype + Critique. Give pairs a minute to generate possibilities before using the three on-screen options. Select a provisional format and connect it to the need. Try the question form as one possible touchpoint, even if anonymous Q&A; is not the selected format.

Facilitate: 1:00 generate; 1:30 compare; 1:30 volunteer task; 1:00 critique and record.

Optional probes: Discuss whom the format might leave out, what the small prototype can teach us, and one revision supported by a reason. **Record:** One format, its rationale, one tradeoff, and one proposed revision.

03 / 15:00-20:00 / CHAPTERS 8-10

What would convince us that the design helps attendees, beyond a usable question form?

Open: Evaluation + Reflection. Use the evaluation choice question to distinguish critique, observation, and inspection. Then propose evidence for the attendee need chosen earlier. A smooth form interaction and a useful career event are different claims.

Facilitate: 0:30 concept check; 2:30 discuss evidence; 1:30 propose a follow-up study; 0:30 name a transfer to visualization.

Optional probes: Specify the participants, task or setting, and an observation that would make us revise the design. Discuss what one classroom volunteer cannot establish. **Record:** One evaluation plan and one method to carry into a visualization project.

A real case, provisional claims

The DVC case supplies decisions to reason about. The reading supplies methods. No attendee research or classroom results are claimed in these materials.

What is known	What remains to investigate
DVC is planning its first career panel. This project is a reading presentation for Major Studio.	Which attendees to prioritize; their current experiences; the panel's goals; access needs; useful formats; and evidence of value.

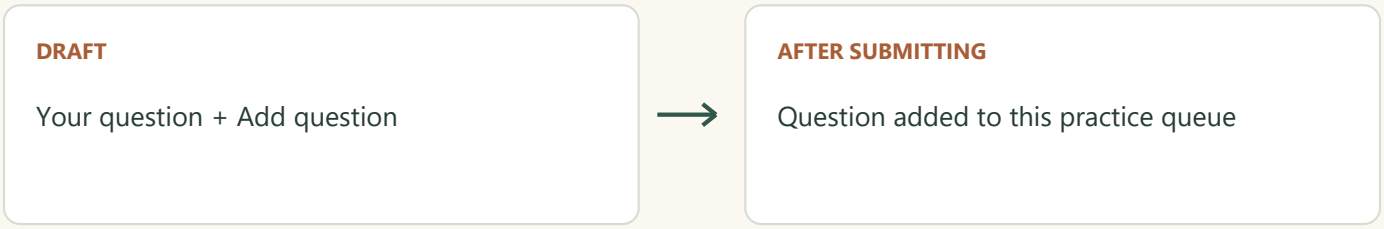
Three candidate needs and formats

Working hypothesis	Format to explore	A tradeoff to discuss
Compare career paths	Several speakers answer shared questions.	Comparable answers may compress different experiences.
Find a first step	Speakers describe decisions and turning points.	A useful story may depend on resources an attendee lacks.
Ask candid questions	A moderated anonymous question queue.	Less public speaking; the moderator still controls selection.

These pairings are examples for discussion, not fixed matches or research findings. Another format, or a revision of the need, may be more appropriate.

The prototype activity

Say to the volunteer: You have a career question but prefer not to speak to the room. Use this page to get your question to the moderator.



Explanatory state sketch based on the submitted app, rather than a screenshot. The demonstration stays on the device and sends nothing to DVC.

Observe: Let the volunteer try without naming a button. Afterward, ask what happened and what they expect next. The form tests one interaction; it does not deliver career advice or test the whole panel format. [6-7, 9]

Connect a claim to its evidence

Use this sheet to keep critique, analytical predictions, and observations distinct. The examples below are proposed activities, not findings from completed studies.

Approach	DVC example	What it can / cannot support
Critique [8]	Discuss why a format serves a goal and whose needs it may overlook.	Offers reasoned perspectives and possible revisions; does not establish actual use.
Empirical evaluation [9]	Observe an intended attendee attempt the question task.	Reveals behavior and task difficulties in that setting; does not by itself prove career value.
Analytical evaluation [10]	Inspect feedback and walk through the task using explicit assumptions about attendees.	Predicts likely difficulties; the predictions depend on the principles and assumptions used.

A possible follow-up for the case

If the working need is finding a first step, a small panel rehearsal could ask intended attendees to explain a concrete next action and why it fits their situation. A later follow-up could explore whether they attempted it and what got in the way. Agreement in class or a successful form submission alone would leave the larger claim open.

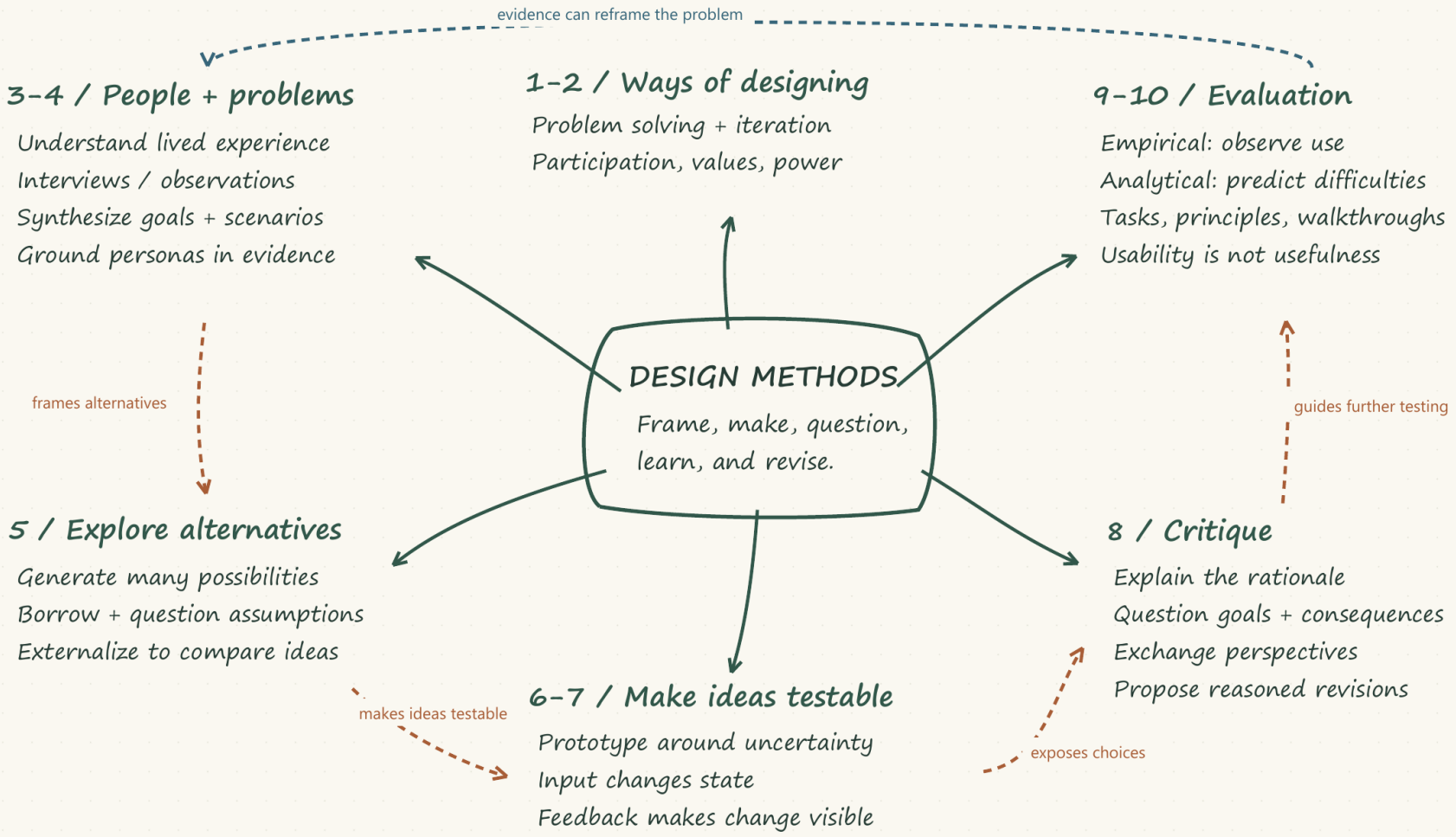
Transfer to MS Data Viz

Case decision	Visualization application
Whose career need matters?	Specify whose question a visualization should help answer, including people absent from the design process.
Which format supports that need?	Compare representations and interaction patterns against a purpose, rather than choosing by appearance alone.
What changes after an action?	Make selections, filters, and their effect on the displayed data visible.
What evidence would justify the design?	Separate comprehension and task observations from preferences, inspection findings, and broader usefulness.

Capture during discussion: Audience + need / format + rationale / concern + revision / evidence to seek. Use the app fields and download class notes from Reflection. Empty fields are acceptable; do not fill them with invented results.

Design Methods: learning through design

Amy J. Ko, 2nd ed. | Concept relationships across chapters 1-10



DVC connection: attendee need -> panel format -> question-form trial -> evidence -> revision

Solid branches organize concepts. Dashed arrows describe selected relationships; iteration can revisit any branch.

Digital-paper drawing with individually placed text and paths. AI-assisted synthesis; not a photographed hand drawing.

Source: Ko, Design Methods, chapters 1-10.

Reading, relationships, resources

All summary language is paraphrased. Numbered references throughout the packet correspond to the chapters below. DVC applications and visualization connections are interpretations developed for this project.

Reading the mind map

The center organizes a practice of learning and revision. Solid branches group concepts; dashed arrows explain selected connections. Understanding people frames alternatives; prototypes expose decisions for critique; evaluation can send us back to the problem. Participation, values, and power influence every branch. This is a selective synthesis, not a prescribed order or a complete inventory of the book.

Primary reading

Amy J. Ko. *Design Methods*, 2nd edition. Online edition, accessed October 30, 2026.
faculty.washington.edu/ajko/books/design-methods/

[1] What designers do

[2] How to design

[3] How to understand problems

[4] How to define problems

[5] How to be creative

[6] How to prototype

[7] How to design interfaces

[8] How to be critical

[9] How to evaluate empirically

[10] How to evaluate analytically

Project materials

Interactive presentation - Shared-screen activities, choices, book sidebar, and the G presenter guide.

Course repository - App source, bilingual guide, and the assignment materials in submission/.

The standalone mind-map PNG can be uploaded with this PDF if a separate image link is requested. The digital-paper PDF preserves the drawing as a separate page. Editable notes accompany the packet.

Preparation and scope

Prepared with AI assistance for synthesis, facilitation planning, wording, code, and the digital-paper map. The map uses individually positioned drawing elements rather than a dedicated mind-map tool. It is not represented as the presenter's hand drawing. No participant data or completed-study results are included. The book and its excerpts retain their original copyright and license.